

Adaptive Augmentation of Medical Data Using Independently Conditional Variational Auto-Encoders

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Abstract—Current deep supervised learning methods typically require large amounts of labeled data for training. Since there is a significant cost associated with clinical data acquisition and labeling, medical datasets used for training these models are relatively small in size. In this paper, we aim to alleviate this limitation by proposing a variational generative model along with an effective data augmentation approach that utilizes the generative model to synthesize data. In our approach, the model learns the probability distribution of image data conditioned on a latent variable and the corresponding labels. The trained model can then be used to synthesize new images for data augmentation. We demonstrate the effectiveness of the approach on two independent clinical datasets consisting of ultrasound images of the spine and magnetic resonance images of the brain. For the spine dataset, a baseline and a residual model achieve an accuracy of 85% and 92%, respectively, using our method compared to 78% and 83% using a conventional training approach for image classification task. For the brain dataset, a baseline and a U-net network achieve an accuracy of 84% and 88%, respectively, in Dice coefficient in tumor segmentation compared to 80% and 83% for the convention training approach.

Index Terms—Data augmentation, deep learning, conditional variational auto-encoder, magnetic resonance, ultrasound, center-line identification, tumor segmentation.

I. INTRODUCTION

DATA-DRIVEN models in medical image analysis utilize machine learning algorithms to learn information about the anatomy from a set of images. Due to the power of these algorithms in learning structures directly from the data, machine learning for medical image analysis is becoming a widespread technology to solve real-world problems in

clinics [1], [2]. In particular, deep models have shown significant success in automatically extracting key features from clinical images and utilizing them for image classification or segmentation [3]–[6]. For instance, Sirinukunwattana *et al.* proposed a spatially constrained convolutional neural network (CNN) architecture for detection of nuclei in histopathology images [7]. Upon detection of the nuclei, they used a neighboring ensemble predictor along with a standard softmax CNN to classify the cancerous cells.

Despite their success in image recognition tasks, deep models require large amounts of data to be regularized. This is required in order to guarantee a minimum difference between train and test errors, which is crucial in clinical applications. However, small sample size is a typical characteristic of medical image datasets since clinical data collection is associated with high complexity and cost. On top of data acquisition, annotation and cleaning is often required prior to training, which adds a significant cost overhead. Moreover, inter-observer interpretation agreement of medical data is highly dependent on clinical expertise [8]–[12], which directly affects the training of models. In order to avoid the costs associated with collecting large datasets, two main approaches have been investigated:

- 1) model engineering and learning strategies, which involve designing models and developing learning algorithms that can learn from small datasets,
- 2) data augmentation, which tries to expand the existing dataset in order to introduce more intra-class variance in the data.

These two approaches have been reviewed in the following.

A. Model Engineering and Learning Strategies

The architecture of deep models is of crucial importance, which directly affects their performance [13]. Deeper architectures do not necessarily correspond to higher accuracies. Therefore, careful design of the model is required to efficiently learn the hidden features of the data. Hence, specific modules such as residual [13], dense [14] and inception [15] were introduced in order to achieve more accurate object recognition. Optimizing deep models for medical image analysis, in particular, is challenging due to the small size of datasets. In a study, Shaikhina *et al.* showed that a large set of various initial parameters (≥ 2000) is required to mitigate the effect of insufficient sample size [1]. Another group of methods leverage the context via down scaled input images.

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Kamnitsas *et al.* proposed a dual path CNN model for brain lesion segmentation that incorporates local and global contextual information via two separate input streams [16]. Ronneberger *et al.* proposed the U-net architecture that utilizes data augmentation in order to overcome the issue of limited data samples [6]. Their architecture outperformed previous state-of-the-art on segmentation of the neural structures in microscopic images. However in order to achieve high performance, the input data needed to be augmented via random transformations and deformations. Since the introduction of the U-net architecture, different variations of the same model have been proposed for various tasks [5], [17], [18]. In a study, Clark *et al.* incorporated the inception reduction blocks from [15] into the U-net model for prostate gland segmentation in diffusion weighted magnetic resonance images [19]. Their proposed method achieved an average Dice score of 0.89.

Aside from engineering the model architecture, learning strategies have been proposed to leverage small amounts of data for training. For instance, active learning algorithms have been investigated for training high performance models with minimum supervision. These algorithms are well-suited to problems, where data annotation is costly. A typical *active learner* starts the training with a subset of the dataset, which is labeled. Then, it selects informative samples, *i.e.*, queries, from unlabeled data and asks an expert to label them, which are added to the training dataset. The model is re-trained on the new set and this procedure is repeated until the model reaches the desired performance.

Active learning has been considered a viable solution to train efficient models for medical image analysis [20]–[23]. In a study, Stanitsas *et al.* utilized active learning to recognize three different types of cancerous tissue, namely, breast cancer, prostate cancer and myometrium [24]. They proposed a multi-stage training method to avoid overfitting, where at each stage a CNN is trained on a small number of labeled samples and then used to predict the class labels of the “active pool.” The predictions are ranked based on an uncertainty measure and then presented to an expert for labeling. Their experiments showed that active learning leads to faster training of the model, while achieving similar performance compared to randomized sampling methods. In another study, Yang *et al.* proposed a similar approach for biomedical image segmentation [25]. They obtained uncertainty and similarity measures from a fully-convolutional network to determine the most representative data samples to annotate. Those samples are then labeled by an expert and added to the training set.

Transfer learning has also been investigated to overcome the problems associated with small size of dataset. Two major strategies have been introduced: 1) using a pre-trained network as a feature extractor, and 2) fine-tuning an existing pre-trained model on medical data for a specific task [2]. Nguyen *et al.* used a combination of pre-trained inception and resnet to extract features from microscopic images [26]. They later concatenated the extracted features and trained a classifier to distinguish normal versus abnormal cells. In a recent work, Yap *et al.* investigated the performance of three conventional deep models, namely patch-based LeNet [27],

U-net and transfer learning fully convolutional-AlexNet (FCN-AlexNet) [28] for breast lesion detection in US images [29]. Their experimental results showed that the transfer learning FCN-AlexNet can achieve high F-measures, *i.e.*, the weighted harmonic mean of recall and precision, (0.90 on average).

Typically, the conventional pre-trained networks have a large number of parameters, which is not efficiently utilized for recognition tasks [30]. Moreover, smaller models are less memory demanding and accordingly, can be more easily deployed on standard computing hardware locally in the clinic. However, due to the limited size of medical datasets, designing efficient models remains a key challenge. Hence, data augmentation techniques, which expand the dataset with realistic data are beneficial.

B. Data Augmentation

Previous research indicates that data augmentation acts as a regularizer and prevents overfitting in deep models [31]. Moreover, such techniques can be used to improve the performance of the model in imbalanced problems, which is common in medical data [32], [33]. Therefore, augmentation methods have been an active area of research in the past few years in the computer vision community [34], [35]. In a study, Wong *et al.* showed that when plausible transformations of data are known, augmentation in data space improves the classification performance [36]. In addition, DeVries and Tyler demonstrated that data augmentation in feature space can effectively improve the classification performance of the model [37].

Cubuk *et al.* proposed a procedure for training deep models based on reinforcement learning (RL) [38]. In order to find the best set of transformations for data augmentation, they quantized the parameter space of all of the known image transformations, *e.g.*, rotation and translation. They used RL to search the parameter space and optimize the parameters based on the performance of the classifier. In another work, Lemley *et al.* introduced a technique, called Smart Augmentation, that generates new images via combining samples from the same class [39].

Variational auto-encoders (VAEs) [40], as a class of generative models that learn an embedding space to approximate the distribution of data, have been utilized for data augmentation [41]. Wan *et al.* showed that VAEs can be helpful in imbalanced problems [42]. They modified the original MNIST dataset [27] by reducing the number of samples in some classes and trained a VAE on the reduced dataset. Then, they generated samples by the trained VAE and augmented the modified dataset with synthetic images. Their experimental results showed that the generated samples contribute to the high classification accuracy.

In addition to VAEs, generative adversarial networks (GANs) have also been investigated for data augmentation. Antoniou *et al.* proposed a model based on GANs, called Data Augmentation Generative Adversarial Network (DAGAN), which learns to transform an input image of a given class to other images of the same class [43]. They showed that DAGAN improves the classification accuracy

on MNIST [27], Omniglot [44] and VGG-Face [45] datasets. Zhu *et al.* proposed a framework based on cycle-consistent GAN [46] and regular CNN to learn unpaired image-to-image translation from reference to target classes. Their experiments show that GAN-based data augmentation techniques can lead to improvement of classification accuracy by 5-10%. Han *et al.* showed that adversarial networks can generate realistic magnetic resonance (MR) images of brain for data augmentation and physician training [47]. Their experiments showed that even an expert physician fails in distinguishing between real and synthetic images in some cases. In another study, Nie *et al.* used GANs to generate computed tomography (CT) images from their corresponding MR images [48].

GANs typically try to match the distribution of source (synthetic) to target (real) domains. However, this can be problematic when the target domain has over or under representation of some classes. A recent study by Cohen *et al.* showed that the generated images by GANs are often misleading for diagnosis [49]. This is due to the fact that in an adversarial setup, the discriminator can be biased by certain features, *e.g.*, tumors in image, and hence, assign more capacity of its weights to learn those features. Accordingly, the generator tries to synthesize samples that contain those features to increase the loss of the discriminator, which leads to an over-represented feature in generated samples. Moreover, GANs usually suffer from instability during training and require improvements [50]–[52].

Conventional augmentation techniques, which include static transformations such as translating the image by a few pixels, mirroring, randomly cropping the image and RGB channel shift, do not consistently improve the performance of deep models across different datasets. For example, while mirroring the image improves the training accuracy on CIFAR-10 dataset [53], it has limited effect on MNIST, due to the different symmetrical features of the images between these datasets [38]. Moreover, although conventional augmentation methods enforce invariance in models, they cannot incorporate prior knowledge during training. Therefore, a method for adaptive data augmentation based on the performance of the discriminative model could be beneficial during training. Hence, the objective of this study is to 1) introduce a flexible generative model for data augmentation that can easily incorporate prior knowledge into the synthetic data and, 2) propose an adaptive algorithm that utilizes the generative model to train the discriminative model.

C. Contributions

In this paper, we propose

- 1) a conditional generative model based on VAEs that learns the embedding space independent of the labels, which is the key difference when compared against the method proposed in [54], and
- 2) an adaptive training algorithm, which utilizes the generative model to augment the training set of a discriminative model.

Conditional variational auto-encoders have been previously investigated [54]. However, by enforcing independence

between the embedding space and the labels, we show that the proposed generative model is capable of incorporating prior knowledge and hence, can generate realistic samples from sparse parts of the distribution of data. In addition, since the embedding space and the labels are independent, we show that the model can generate samples with different features while preserving the label correspondence. Using this property, we demonstrate that a discriminative model can be trained on an augmented dataset, where the proposed VAE can adaptively generate samples, which contribute to the training accuracy of the discriminator. Since our proposed VAE is not adversarially influenced by the discriminator, it does not over- or under-represent certain features of images.

In order to investigate the impact of the adaptive training framework on the performance of a discriminative model, we consider the following clinical problems:

- identification of the midline of the spine in transverse ultrasound (US) images,
- segmentation of the whole tumor in FLAIR MR images of brain.

We show that the proposed adaptive training improves the performance of deep discriminative models.

II. METHODS AND MATERIALS

We start with the preliminaries followed by a brief discussion about the background of the variational auto-encoders. We, then, proceed to our proposed model and derive the variational lower bound for it. Next, we discuss sampling from the learned distribution and introduce transformations, which are used to enforce the prior knowledge into sampling. We then use the generative model along with the transformations as building blocks of a training algorithm and introduce an adaptive method to train a discriminative model.

A. Preliminaries and Notation

In order to formulate the problem of adaptive data augmentation, let (x, y) be a sample pair from dataset $\mathbf{D}$, where x is the input and y is its corresponding label. Also, let z be a latent variable in an embedding space with a known distribution parameterized by ξ . Our goal is to design a generative model that encodes the appearance features of x into z , independent of its label. Later, it draws samples from the embedding space and uses y to generate new synthetic data, where the correspondence between the input and labels is preserved. Let $\mathcal{G}$ be the generator of the model, with two inputs: a random sample from the latent space (z) and y . Also, let $\mathcal{M}$ be a discriminative model with parameter set ψ .

We hypothesize that $\mathcal{G}$ along with a transformation function that modifies the labels ($T(y)$), can be used to adaptively augment the $\mathbf{D}$ during training of $\mathcal{M}$, based on its performance. $T(y)$ is a heuristic, which incorporates the prior knowledge about $\mathbf{D}$ into labels. Hence, it can modify the distribution of the training labels in order to match that of the test data. Moreover, since the appearance features of the input are encoded into z , different samples from the embedding space correspond to various patterns in the synthetic data. Therefore, $\mathbf{D}$ can be adaptively augmented with respect to the performance of $\mathcal{M}$.

during training. This is feasible by changing the parameters of the distribution of z , based on the training loss of $\mathcal{M}$. Accordingly, a set of synthetic data that contribute to the performance of $\mathcal{M}$ is added to $\mathbf{D}$.

B. Variational Auto-Encoder

Variational auto-encoders (VAEs) are a type of directed graphical models with parametric latent variables, such as Gaussian latent variables. VAEs are comprised of a variational approximator $q_\phi(z|x)$, which tries to estimate the intractable posterior p , and a generative model $p_\theta(z)p_\theta(x|z)$, where z is the latent variable and x is the input. The approximator and the generator are parameterized by ϕ and θ , respectively. The goal of the VAE model is to jointly learn θ and ϕ . This can be efficiently achieved via stochastic gradient variational Bayes (SGVB) framework [40], where the variational lower bound is maximized with respect to the variational and generator parameters (ϕ and θ). In a conventional VAE model with embeddings z , the variational lower bound is defined as:

$$\mathcal{L}_{\text{VAE}}(\theta, \phi, x) = -KL(q_\phi(z|x) \| p(z)) + \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)], \quad (1)$$

where $KL(q \| p)$ is the Kullback-Leibler divergence of q and p . Since the gradient of $\mathcal{L}$ with respect to ϕ is not straightforward to calculate, $q_\phi(z|x)$ can be reparameterized by a known differentiable function $g_\phi(x, \epsilon)$, where ϵ is a random noise with prior $p(\epsilon)$. The reparameterization trick allows the variational lower bound to be optimized using gradient descent. Both the variational and the generator parts of the VAE can be modeled via a neural network.

C. Independent Conditional VAE

Let $p(x, y, z)$ be the joint distribution of the input, labels and the latent variable. The goal of the independent conditional VAE (ICVAE) is to learn $p(x, y, z)$ and later, generate images for data augmentation through sampling from the sparse parts of the distribution. We hypothesize that the biased sampling contributes to the training of a discriminative model, due to more balanced training data, as well as increased intra-class variance in each batch.

Learning to encode structures in the embedding space in the VAE model has been investigated in [54]. However, the existing conditional VAE encodes both the labels and the input into the embedding space.

For data augmentation, it is preferred to have the embedding space and the labels independent of each other. This allows one to control the characteristics of the image independently and also incorporate prior knowledge into the labels in order to balance the data. If y and z are independent, by conditioning over z , $p(x, y, z)$ can be rewritten as:

$$p(x, y, z) = p(x|y, z)p(y|z)p(z) = p(x|y, z)p(y)p(z). \quad (2)$$

Similar to a conventional VAE, we can consider a known prior distribution for $p(z)$, e.g., $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Moreover, we can use $p(y)$ to impose prior knowledge about the dataset into sampling.

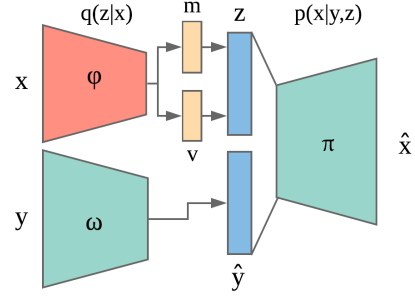


Fig. 1. Block diagram of the independent conditional VAE (ICVAE) model. The variational approximator and the generator are parameterized by ϕ (red) and $\{\omega, \pi\}$ (green), respectively. After training, synthetic data can be generated from a random latent sample z and a particular label y .

In the ICVAE model, $p(x|y, z)$ and accordingly, the joint distribution are parameterized via θ . Therefore, since $p(z)$ and $p(y)$ are known, the joint distribution can be learned by optimizing θ . In order to enforce the embedding and label independence, the variational and the generator models have to be modified such that y has no effect on z . Figure 1 shows a block diagram of the ICVAE model, which consists of a variational (red), a generator and an encoder model (green) parameterized by ϕ , ω and π , respectively. The variational approximator encodes the input into the latent variable z and the generator uses the labels y and the latent variable to reconstruct the input $\hat{x}$. Accordingly, the generator would approximate $p(x|y, z)$ after training. In the ICVAE model, the latent variable and the labels are concatenated prior to the first layer of the generator. Since they may have different dimensionalities, the reconstructed input $\hat{x}$ would be biased by the one with higher dimensionality. This is due to the dominance of the vector with higher dimensionality in weighted summation in a fully connected layer, or convolutional feature maps in a convolutional layer. In order to balance the impact of y and z on $\hat{x}$, an encoder parameterized by ω compresses y into a vector $\hat{y}$ with matching dimension with z . Hence, $p(x|y, z)$ is parameterized by $\theta = \{\omega, \pi\}$:

$$p_{\pi, \omega}(x, y, z) = p_{\pi, \omega}(x|y, z)p(y)p(z) \quad (3)$$

1) **Variational Lower Bound:** In order to learn $p_{\pi, \omega}(x|y, z)$, we use the SGVB framework to optimize π and ω . Let $p(z|x)$ be the true posterior, which is going to be approximated by $q(z|x)$ through minimizing their KL divergence. It can be shown that the KL divergence for the ICVAE model can be written as (see Appendix A):

$$KL(q(z|x) \| p(z|x)) = \log p(x) - \sum_z q(z|x) \times \log \frac{p(x, y, z)}{p(y|z, x)q(z|x)} \quad (4)$$

ICVAE's objective is very similar to the conventional VAE. By maximizing the likelihood of the data, the ICVAE model is enforced to first, learn a mapping from the distribution of the data to a known parametric prior ($p(z)$) and then, use samples from the latent space along with the label information to generate realistic synthetic data.

Algorithm 1 AdaTrain

Input: $\mathbf{D} = \{(x, y)\}, \mathcal{G}, T(y), \mathbf{E} = \{e_i\}, \mathcal{M}_\psi, \mu_0, \sigma_0$
Output: $\mathcal{M}_\psi$ with trained ψ
 $\zeta \leftarrow (\mu_0, \sigma_0)$
for e in $\mathbf{E}$ **do**
 $z \leftarrow \text{Sample}(\mathcal{N}(\zeta))$
 $\mathbf{D} = \mathbf{D} \cup \{\mathcal{G}(z, T(y_D))\}$
 for each minibatch $\mathcal{B}$ in $\mathbf{D}$ **do**
 $g \leftarrow \text{grad}(\mathcal{L}_D(\mathcal{B}))$
 $\psi \leftarrow \text{SGD}(g, \psi)$
 end for
 $\bar{\ell} = \frac{1}{N} \sum_i \mathcal{L}_D(\mathcal{G}(z^i, T(y_D^i)))$
 $\beta \leftarrow \{z^i : \mathcal{L}_D(\mathcal{G}(z^i, T(y_D^i))) > \bar{\ell}\}$
 $\zeta \leftarrow (\text{mean}(\beta), \text{std}(\beta))$
end for

With the use of the Bayes law, the variational lower bound can be rewritten as (see Appendix B):

$$\begin{aligned} \mathcal{L}_{\text{ICVAE}}(\phi, \pi, \omega, x, y) = & -KL(q_\phi(z|x) \| p(z)) \\ & + \mathbb{E}_{q_\phi(z|x)} [\log p_{\pi, \omega}(x|y, z)] \\ & + \log p(y) - \log p(y|x). \end{aligned} \quad (5)$$

$\log p(y) - \log p(y|x)$ is negligible, if the batch size is large (see Appendix C).

The ICVAE model is trained using the AEVB algorithm, introduced in [40], with $L = 4$ random samples from the noise distribution $p(\epsilon)$ for each data point in the batch.

D. Adaptive Training Using the ICVAE Model

As shown in equation (2), $p(x|y, z)$ is directly proportional to $p(y)$ and $p(z)$: $p(x|y, z) = \frac{p(x, y, z)}{p(y)p(z)}$. Therefore, any changes to the distributions of z and y are going to affect $p(x|y, z)$ and accordingly, the generated image. Let $\mathbf{Y} = \{y' : p(y') < \tau\}$ represent the set of labels from the original training dataset, where their probability is smaller than a threshold τ . Moreover, let $\zeta_0 = \{\mu_0, \sigma_0\}$ be the initial parameters of the probability distribution of z . In a trained ICVAE, ζ can be tuned based on the loss of the discriminator ($\mathcal{L}_D$).

In addition, in the case of imbalanced data, drawing samples from $\mathbf{Y}$ and generating their corresponding images would balance each training batch. Since z and y are independent, labels and the distribution parameters of the embedding space have different effects on the generated images. While ζ controls the characteristics of the image, $y' \in \mathbf{Y}$ would lead to a more balanced discriminative model $\mathcal{M}$.

The goal of training via adaptive data augmentation (AdaTrain) is to utilize a trained ICVAE $\mathcal{G}$, adjust ζ based on the discriminative loss, and, if required, utilize y 's and generate corresponding images. The synthetic images are then augmented with the training batch. Algorithm 1 summarizes the AdaTrain. Sampling from $\mathbf{Y}$ can be done via a transformation function $T(y)$ that modifies the distribution of the training labels based on the prior knowledge about the data. In order to adjust ζ , AdaTrain uses the initial ζ_0 to augment the training data for a certain number of epochs, which are referred

to as “rounds.” After the first round, it evaluates $\mathcal{L}_D$ for all of the generated images. Then, it updates ζ with the mean and the standard deviation of the latent variable values, whose corresponding images indicated higher discriminative loss compared to the mean loss of the batch. Therefore, in each round, AdaTrain augments the original dataset with synthetic difficult samples that the discriminator was not accurate in identifying their key features. Note that the number of epochs in each round needs to be sufficiently large so that the discriminator has enough number of iterations for learning. $\mathbf{E} = \{e_i\}$ represents the set of augmentation rounds, where e_i indicates the number of epochs for the i 'th round.

E. Datasets

Table I summarizes the datasets used to evaluate the performance of the proposed method. We consider the following clinical problems and discuss the details of the respective datasets:

A) identification of the center-line of the spine through classification of spinal transverse ultrasound (US) images,

B) segmentation of the tumor in T2 Fluid Attenuated Inversion Recovery (FLAIR) magnetic resonance (MR) scans of brain.

1) *Dataset A-Transverse US Images of the Spine*: After obtaining written consent, twenty volunteers aged between 20 and 40, whose body mass indices (BMI) were less than 30, participated in the study (ethics certificate number: H15-01310). From each volunteer, an expert sonographer acquired three separate 2D US sequences from the right, left, and the center line of the spine, each starting from the sacrum to the T12 (60 US sequences in total). The sequences were reviewed by the sonographer for quality control and then anonymized prior to saving. In order to maintain the scan consistency, the following protocol was defined:

- A frame is considered “center-line,” if the center of the transducer is within the width of the spinous process.
- Prior to each scan, the sonographer identifies the sacrum as the starting point.
- The off-center scans are captured within an arbitrary distance, ranging from 1 cm to 6 cm, from the center line.

Due to high inter-frame correlation, the sequences were subsampled every 5 frames. All of the frames were divided into two categories for classification, *i.e.*, center-line and off-center frames. Sixteen volunteers were randomly selected and their corresponding data were considered for training and validation, which consists of 11240 off-center and 5610 center-line frames (16850 in total). Out of 16850 frames, 500 were randomly selected for training the ICVAE and the classification models. The data from remaining 4 volunteers were put into the test set, containing 11720 images. In order to evaluate the performance of the classifier, the test data were balanced by manually selecting an equal number of center-line and off-center images from the acquired set. The images of both train and test sets were resized to 102×136 for memory considerations.

The ultrasound images were presented to an expert radiologist for second annotation without any exposure to the

TABLE I

SUMMARY OF THE DATASETS USED IN THE STUDY. THE PERFORMANCE OF THE PROPOSED METHOD IS EVALUATED ON ULTRASOUND IMAGES OF THE SPINE AND AXIAL PLANES OF BRAIN MR SCANS.

Dataset	Modality	Category	Train images	Test images	Image size	Public
A-Transverse images of the spine	Ultrasound	Classification	500	11720	102×136	No
B-FLAIR volumes of the brain	MR	Segmentation (2D)	3000	29550	128×128	Yes

previous labels. She annotated the center-line of the spine in each ultrasound frame. For a given image, if the annotated center-line falls within the width of the spinous process at the center of the image, the frame is marked as “center-line,” otherwise, “off-center.” The center-line and off-center images had 92% and 100% annotation match between the first and the second set of labels, respectively.

All of the sequences were collected by a C5-2/60 curvilinear transducer operating at 3.3 MHz attached to a SonixTOUCH US machine (Analogic Corp. Richmond, Canada). Data acquisition parameters including depth, gain, and focal point were optimized by the sonographer for each volunteer.

2) *Dataset B-MR Volumes of the Brain*: In order to investigate the impact of the proposed method on image segmentation, we use the publicly available brain tumor segmentation (BraTS) dataset [55], [56]. The data involve clinically-acquired pre-operative multimodal MR scans of glioblastoma and lower grade glioma from 19 institutes across the world. The scans describe the native T1-weighted, post-contrast T1-weighted, T2-weighted and FLAIR (fluid-attenuated inversion recovery) volumes. Following a consistent protocol, the volumes have been segmented manually by one to four raters. Later, the annotations were reviewed by experienced neuro-radiologists. Annotations include regions of the whole tumor, the peritumoral edema, and the necrotic and non-enhancing tumor core. All of the volumes are co-registered to the same anatomical template, interpolated to the same resolution (1 mm^3) and skull-stripped. For this study, high grade glioblastoma FLAIR scans and their corresponding whole tumor labels are used. Nineteen volumes out of 211 were randomly selected as training and the remaining 192 were considered as test data. The axial planes of each volume were resized to 128×128 pixels and used for 2D segmentation of the cancerous tissue.

3) *Significance*: Image classification and segmentation are two common sub-categories of medical image analysis. These problems are significant in clinics, since they are often used in intermediate steps for diagnostic or therapeutic procedures. Therefore, accuracy is of crucial importance for diagnosis or treatment and depends upon application.

In particular, transverse US images of the spine can visualize the spinal anatomy to identify the needle puncture site for spinal anesthesia. Hence, inaccurate identification of the center-line of the spine leads to multiple needle insertion attempts that increase the procedure time, excessive pain and patient anxiety [57]. The level of accuracy here is determined by the width of the interspinous ligament, through which the needle should pass. According to the literature, the width of the ligamentum flavum is approximately 7 mm [58]. Also, glioblastoma is one of the most common and most aggressive

types of brain tumor, where the median survival time is 15 months [59]. Hence, early detection of its biomarkers is advantageous for therapy in a timely manner [60]. However, due to their heterogeneity and inherent shape variability, segmentation of brain tumors in MRI scans remains a challenge [61].

F. Transformation Functions

The transformation function (T) can be defined with respect to the prior knowledge about each particular dataset.

1) *Dataset A*: Since the off-center images consist of both the left and right scans, the training data contain considerably fewer center-line frames compared to off-center (119 vs. 381). Hence, the dataset is imbalanced. Therefore assuming that $y = 1$ denotes the label for the center-line images, the following transformation function is proposed to balance the training data:

$$T_A(y) = Y \sim B(|\{y\}|, 1 - \frac{|\{y : y = 1\}|}{|\{y\}|}), \quad (6)$$

where $B(n, p)$ is the binomial distribution with n total experiments, with the success probability, *i.e.*, center-line label of p . Y is a random sample from $B(n, p)$.

2) *Dataset B*: As shown in Figure 2, the majority of the axial planes of the training set do not contain any traces of the tumor. In order to balance the labels, the training batches need to be augmented with the synthetic data that indicate glioblastoma. Therefore, the transformation function should alter the shape of the tumor label. This is achieved by applying a random elastic deformation on tumor labels:

$$T_B(y) = DF(y) \quad \forall y \in \{y_i : \sum y_i > t\}, \quad (7)$$

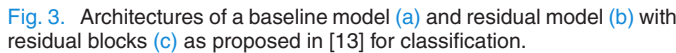
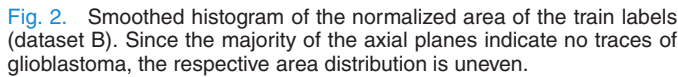
where DF denotes elastic deformation [62], which slightly alters the shape of the cross-section of the tumor and t is the area threshold on the binary label y .

G. Implementation

The proposed augmentation algorithm and the independent conditional VAE were implemented in Theano (code publicly available at <https://github.com/mpslxz/icvae> and https://github.com/mpslxz/theano_ops). The models were trained on a GeForce(R) GTX 980 Ti GPU with 6 GB of memory. The host machine was running on a Core i7 CPU with 16 GB of RAM.

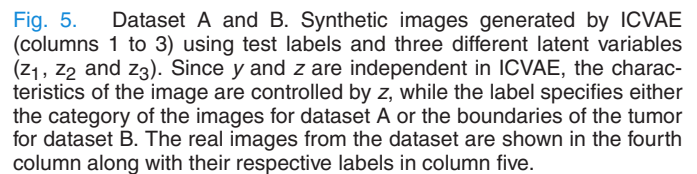
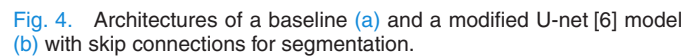
III. EXPERIMENTS AND RESULTS

The performance of the proposed method was evaluated on datasets A and B.



A. Model Architectures

1) **ICVAE**: The ICVAE model uses two fully connected layers with 1024 and 512 neurons symmetrically in the variational approximator and the generator. The nonlinear activation function for each layer of the generator and approximator is *tanh* and the output is activated via the *sigmoid* function. The dimension of the latent variable is set to 10. Moreover, the encoder of the labels uses one fully connected layer with *sigmoid* activation, which encodes the labels into a vector with the dimension of the latent variable. For consistency, this architecture is maintained throughout all of the experiments. Figure 5 shows synthetic data generated by ICVAE with



2) **Classification:** The baseline discriminative model for classification of dataset A uses a convolutional neural network with 4 layers, where each layer has 64 kernels with size 5×5 (Figure 3-a). After each layer, a 2×2 maxpooling operation is performed. The last convolutional layer is followed by a fully connected layer, which converts the final feature maps into class predictions using a *softmax* function. Moreover, a model with residual blocks is developed for classification (see Figure 3-b). It utilizes four convolutional layers with 16 kernels of size 5×5 , followed by a residual block, which includes two consecutive convolutions, each with 8 kernels of

size 3×3 . Similar to the baseline model, the final feature maps are converted to class probabilities using a fully connected layer and a *softmax* function. Rectified linear units (ReLU) [64] were used as nonlinear activations. In addition, all of the feature maps are randomly masked by 0 via dropout operation [65] with probability 0.25. The weights of the models were optimized by minimizing the negative log-likelihood of the outputs given targets.

3) Segmentation: The baseline segmentation model is an end-to-end conv-deconv network that maps the input image to its corresponding label. Both the conv and the deconv paths have 4 convolution/deconvolution layers with 32, 16, 16, and 32 kernels of size 3×3 . After the final convolutional layer, a fully connected layer with 128 neurons connects the conv to the deconv network. Similar to the classification model, each convolutional layer is proceeded by a 2×2 maxpooling operation. Deconvolution is performed via transposed convolution as proposed in [66] and is followed by a 2×2 up-sampling operation in the baseline model. In addition to the baseline model, a neural network based on U-net [6] is developed. The model uses 4 convolutional blocks in the encoding path, each of which performs two consecutive convolution operations with 16, 32, 64 and 128 kernels of size 3×3 , respectively. In the decoding path, each convolutional block is followed by a 2×2 up-sampling and a 2×2 regular convolution operation, as suggested in [6]. Moreover, the feature maps of the encoder are merged with those of the decoder. A fully connected layer with 512 neurons connects the encoding and decoding paths of the model. All of the activation functions of both models are rectified linear units (ReLU) except the output activation, which is the *sigmoid*. During training, the networks minimized the loss function based on Dice coefficient between the outputs and targets, which is defined as $1 - \frac{2 \sum y_i t_i + \epsilon}{\sum y_i + \sum t_i + \epsilon}$, where y and t are network outputs and targets, respectively. ϵ is a smoothing coefficient to avoid undefined loss, when both target and output are zero.

B. Evaluation Protocol

In order to investigate the effect of adaptive data augmentation, the performance of the discriminative models was evaluated in three scenarios:

- 1) trained on the original data (S1),
- 2) trained on the statically augmented data, without tuning ξ (S2), and
- 3) trained on the augmented data with adaptive ξ using AdaTrain (S3).

The classification performance is evaluated via the following metrics:

- classification accuracy (ACC): the correct classification rate for both center-line and off-center classes,
- sensitivity (SNS): the correct classification rate for center-line images,
- specificity (SPC): the correct classification rate for off-center images,
- false positive rate (FPR): the rate of misclassified off-center images,
- false negative rate (FNR): the rate of misclassified center-line images, and

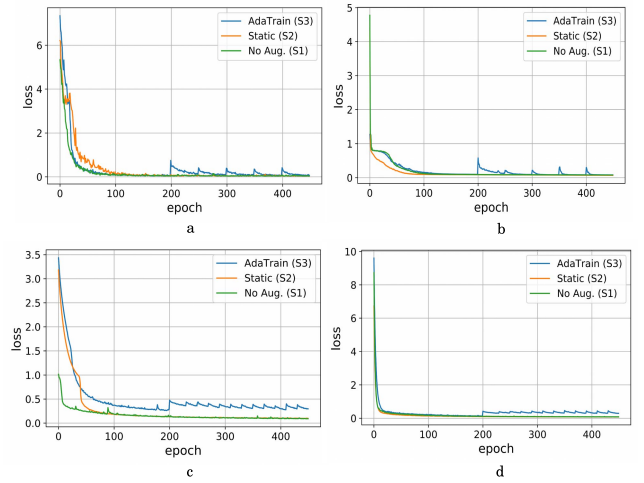


Fig. 6. Training discriminative loss of the baseline and the residual classification models (a and b) and baseline and U-net segmentation models (c and d) for the three training scenarios. In each round of augmentation, ICVAE generates images that the models indicated greater loss than the average. Hence, there is an increase at the beginning of the augmentation rounds. However after several epochs, the models learn new features associated with the augmented data and therefore, the loss values start to decrease.

- the area under ROC curve (AUC) that defines the probability of correct classification for a given class.

The performance of the segmentation model is evaluated using:

- the Dice similarity coefficient [67],
- the Hausdorff distance [68],
- pixel sensitivity, specificity, false positive rate, and false negative rate.

The pixel related metrics have similar definitions to their classification counterparts, except each metric is calculated based on whether or not a pixel belongs to the target region in segmentation map. Tables II and III summarize the experimental results, which are discussed in detail in the following.

C. Dataset A, Classification Performance

1) Original Data: The baseline and residual networks described in section III-A.2 were trained on the original dataset for 450 epochs with batch size 64, using the stochastic gradient descent (SGD) algorithm with learning rate 10^{-4} . The ℓ_2 norm of the weights was added to the negative log-likelihood loss with coefficient $\lambda = 10^{-4}$. After training, the baseline model accurately classified 78% of the test data with 66% and 91% sensitivity and specificity, respectively. Since the training dataset is imbalanced, the model is biased towards off-centered images, hence, low sensitivity. Moreover, the model indicated 10% and 52% false positive and false negative rates, respectively. The residual model classified 83% of the test data correctly with 76% sensitivity and 89% specificity. The false positive and false negative rates of the model were 11% and 30%, respectively. This further shows that the networks cannot identify the center-line images as accurately as the off-center frames.

2) Static Augmentation: After training the ICVAE on the original dataset, the generator along with the transformation

TABLE II

COMPARISON OF THE CLASSIFICATION METRICS OF THE TEST DATA A FOR A BASELINE DISCRIMINATIVE MODEL TRAINED WITH THE ORIGINAL, STATICALLY AUGMENTED AND ADAPTIVELY AUGMENTED DATA. SINCE BOTH THE CENTER-LINE (C) AND OFF-CENTER (O) CLASSES ARE BALANCED IN THE AUGMENTED DATASETS, THE RESPECTIVE MODEL INDICATES HIGHER SENSITIVITY COMPARED TO THAT OF THE ORIGINAL DATA. MOREOVER, ADAPTIVE ADJUSTMENT OF THE LATENT SPACE LEADS TO IMPROVEMENT IN CLASSIFICATION ACCURACY (MCNEMAR'S TEST P-VALUE < 0.0005)

Disc. model	Exp.	Gen. model	ACC	SNS	SPC	FPR	FNR	AUC _c	AUC _o
Baseline	No Aug	N/A	0.78	0.66	0.91	0.10	0.52	0.90	0.88
	Static	CVAE	0.80	0.73	0.85	0.16	0.36	0.88	0.88
		ICVAE	0.81	0.79	0.82	0.22	0.27	0.88	0.88
	AdaTrain	CVAE	0.83	0.75	0.90	0.11	0.32	0.93	0.92
		ICVAE	0.85	0.88	0.81	0.22	0.14	0.93	0.93
Residual	No Aug	N/A	0.83	0.76	0.89	0.11	0.30	0.92	0.92
	Static	CVAE	0.87	0.90	0.83	0.19	0.09	0.94	0.94
		ICVAE	0.87	0.90	0.83	0.19	0.10	0.94	0.94
	AdaTrain	CVAE	0.92	0.94	0.90	0.10	0.05	0.95	0.95
		ICVAE	0.92	0.92	0.92	0.08	0.08	0.95	0.95

TABLE III

DATASET B. COMPARISON OF THE SEGMENTATION METRICS OF THE TEST DATA B FOR BASELINE AND U-NET SEGMENTATION NETWORK TRAINED WITH THE ORIGINAL, STATICALLY AUGMENTED, AND ADAPTIVELY AUGMENTED DATA. WHILE DATA AUGMENTATION VIA ICVAE GENERALLY IMPROVES THE DICE COEFFICIENT, ADAPTIVELY TUNING ξ LEADS TO FURTHER IMPROVEMENT OF THE PERFORMANCE. MOREOVER, THE MODELS TRAINED VIA ADATRAIN SHOW SMALLER STANDARD DEVIATION OF THE DICE COEFFICIENT AND HAUSDORFF DISTANCE

Disc. model	Exp.	Gen. model	Dice $\pm std$	Hausdorff $\pm std$ (mm)	P-SNS	P-SPC	P-FPR ($\times 10^{-2}$)	P-FNR
Baseline	No Aug	N/A	0.80 $\pm$ 0.33	2.78 $\pm$ 3.24	0.83	0.997	0.20	0.74
	Static	CVAE	0.76 $\pm$ 0.35	3.02 $\pm$ 3.4	0.82	0.997	0.27	0.75
		ICVAE	0.82 $\pm$ 0.33	2.78 $\pm$ 3.1	0.85	0.998	0.18	0.72
	AdaTrain	CVAE	0.75 $\pm$ 0.35	3.02 $\pm$ 3.28	0.86	0.997	0.28	0.70
		ICVAE	0.84 $\pm$ 0.33	2.72 $\pm$ 3	0.86	0.998	0.16	0.71
U-net	No Aug	N/A	0.83 $\pm$ 0.30	2.6 $\pm$ 3.02	0.87	0.998	0.17	0.70
	Static	CVAE	0.82 $\pm$ 0.32	2.6 $\pm$ 2.9	0.88	0.998	0.13	0.69
		ICVAE	0.84 $\pm$ 0.30	2.42 $\pm$ 2.72	0.90	0.998	0.10	0.67
	AdaTrain	CVAE	0.81 $\pm$ 0.33	2.76 $\pm$ 3.12	0.88	0.997	0.20	0.69
		ICVAE	0.88 $\pm$ 0.26	2.16 $\pm$ 2.6	0.90	0.999	0.08	0.67

function (6) was used to augment the dataset. z was sampled from $\mathcal{N}(0, 2)$. Using the generator, 500 images were added to the original dataset to balance the number of center-line and off-center images. Later, the same classification models were trained for 450 epochs on the augmented dataset with $\lambda = 10^{-4}$ and learning rate 10^{-4} . The baseline and residual models successfully classified 81% and 87% of the test data, respectively. Moreover, the residual model was 90% sensitive and 83% specific, compared to the baseline network, which indicated 79% sensitivity and 82% specificity. Since the training data are balanced after augmentation between classes, the FPR of both models is closer to the respective FNR (22% vs. 27% for the baseline and 19% vs. 10% for the residual model) when compared against those in the S1.

3) Adaptive Augmentation: The generator of the trained ICVAE was used to adaptively augment each batch of the dataset via algorithm 1. Both classifiers were trained on the original data during the first 200 epochs. Later, the training dataset was augmented in 5 rounds with 50 epochs in each round (450 epochs in total). Similar to static augmentation, initially ξ was set to $(\mu_0 = 0, \sigma_0 = 2)$ and the same transformation function was used to balance each batch in a given round. After 450 epochs, the trained baseline model successfully classified 85% of the test data with 88% sensitivity and 81% specificity, while the residual network correctly classified 92% of the data, being 92% sensitive and specific. FPR and

FNR were 22% and 14%, respectively for the baseline and 8% and 8%, respectively for the residual network. McNemar's significance test [69] along with Bonferroni correction [70], [71] showed that AdaTrain significantly improves the classification performance of both models (p-value < 0.0005).

D. Dataset B, Segmentation Performance

1) Original Data: The segmentation networks, described in section III-A.3 were trained on the axial FLAIR planes of the brain for 450 epochs with batch size 32 and learning rate 10^{-4} . To regularize the models, the ℓ_2 norm of the weights with coefficient $\lambda = 10^{-5}$ was added to the loss function. The average Dice score of the baseline model on the test data was 80% with 83% pixel sensitivity, whereas U-net achieved an average score of 83% with 87% pixel sensitivity on the test set. The baseline model and U-net indicated an average Hausdorff distance of 2.78 mm and 2.6 mm, respectively.

2) Static Augmentation: In order to augment the dataset, random tumor labels were selected from the training data. Using the transformation function (7), the labels were randomly deformed and their corresponding images were generated by ICVAE with $\xi = (\mu_0 = 0, \sigma_0 = 2)$ and were added to the original train set. This procedure was repeated for 2945 random deformations, leading to a dataset of 5890 images in total. After training the discriminative networks on the augmented

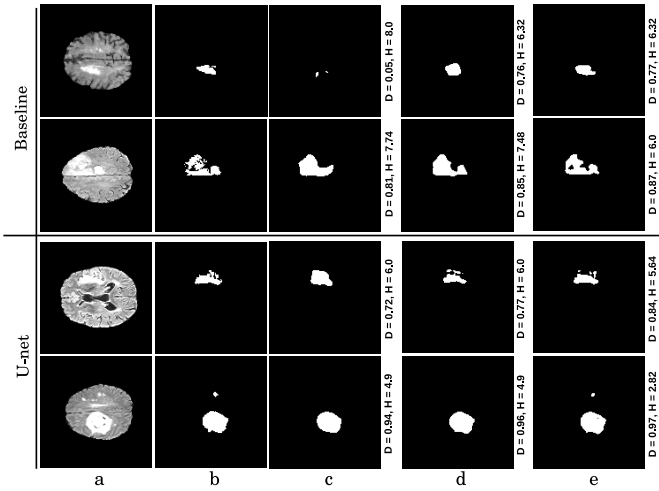


Fig. 7. Dataset B, sample axial FLAIR planes of the brain (column a) and their corresponding tumor ground-truths (column b) along with segmentation results of the discriminative models trained on the original data (column c) versus trained via static augmentation (column d) and AdaTrain (column e). As indicated by the Dice coefficients (D) and Hausdorff distances (H), adaptive data augmentation improves the accuracy of the models on the unseen data.

dataset for 450 iterations with the aforementioned hyper-parameters, the baseline and U-net models reached 82% and 84% Dice coefficient on average on the test data with 2.78 mm and 2.42 mm average Hausdorff distance, respectively. Also, the baseline model indicated an average rate of 85% sensitivity, compared to U-net, which was 90% sensitive.

3) Adaptive Augmentation: The segmentation networks were trained for 200 epochs on the original data and 12 rounds of augmentation via AdaTrain. The first round consisted of 30 epochs, while the others had 20 epochs each. For consistency, initial (ξ) was set to ($\mu_0 = 0, \sigma_0 = 2$) and $T_B(y)$ was used to transform the training labels. The baseline network was then tested on the same unseen data and achieved 84% mean Dice score with 86% pixel sensitivity and 2.72 mm average Hausdorff distance. U-net segmented the tumor with 88% Dice score and 90% pixel sensitivity with 2.16 mm average Hausdorff distance.

The segmentation results of the networks in each scenario have been shown in Figure 7 for sample test images. Note that the models trained with AdaTrain, has lower segmentation error, as is shown by the higher Dice coefficients and lower Hausdorff distances. The Friedman non-parametric significance test [72] on Dice coefficients and Hausdorff distances indicates that AdaTrain significantly improves the segmentation performance of the baseline model and U-net (p-value < 0.0005).

E. Comparative Study

In order to investigate the impact of independence between the appearance features and the labels on data augmentation, the proposed model was compared against a conditional variational auto-encoder (CVAE) [54]. Hence, a CVAE model with the same network architecture as the proposed ICVAE was implemented and trained on datasets A and B

using the same training set as ICVAE. CVAE was trained to maximize the conditional likelihood of the data, as proposed by Mishra *et al.* [73]. All of the training hyper-parameters remained unchanged from the ICVAE model.

CVAE was used for static and adaptive data augmentation (S2 and S3) and the classification and segmentation performance of the discriminative models were investigated after training. Tables II and III summarize the comparison. As indicated in the tables, the classification performance of the baseline and the residual models is comparable when trained on augmented data generated via CVAE and ICVAE. However on dataset B, the segmentation performance of the models is reduced when they are trained on synthetic data, generated by CVAE.

IV. DISCUSSION

In order for ICVAE to approximate $p(x|y, z)$ accurately, the training data should have various samples that represent $p(x, y, z)$. Moreover, since one projection of $p(x, y, z)$ is over labels y , efficient transformation functions are required to accurately transform the existing labels for data synthesis. For classification, $T(y)$ is usually straightforward, since the labels are one-hot encoded vectors that indicate the corresponding class. However, for segmentation problems T depends on anatomical factors such as rigidness, size and shape of the target. Therefore, defining $T(y)$ is a heuristic technique, which requires prior knowledge about the particular training and test data in each study.

Since data synthesis via ICVAE involves random sampling from a probability distribution ($p(z)$), the generated samples might lack expressive features and semantic information. This is significant in high resolution and highly detailed input data. Several methods have been proposed to work around this problem, one of which is training the VAE adversarially [74]–[77]. Another approach is to modify the objective of the model so that it is more sensitive to high frequency and semantic information in the input. Lamb *et al.* proposed to add a discriminative loss as a regularizer to the variational lower bound [78]. Also, Hou *et al.* enforced consistency between the hidden representations of the input and its reconstruction by comparing their corresponding feature maps and penalizing the model for large differences [79]. In our experiments, a relatively simple network architecture could approximate the distribution of the data. However, for more complex and high resolution datasets, applying the proposed solutions for detailed data synthesis to the ICVAE model might be required. Since the latent space of CVAE contains information from both the image and the structure, *i.e.*, the labels, random samples from the latent space with deformed labels do not yield realistic images (see Figure 8). This is particularly critical for segmentation, since the dimensionality of the label is equal to the input. However, because the classification labels have considerably lower dimensionality, when compared to the input (approx. 12500 times less), CVAE's embedding space is essentially dependent on the input. This hypothesis was verified by calculating the ratio of the sum of absolute value of the trained weights between the input and the label in the first layer of the model. For classification (dataset A), the ratio

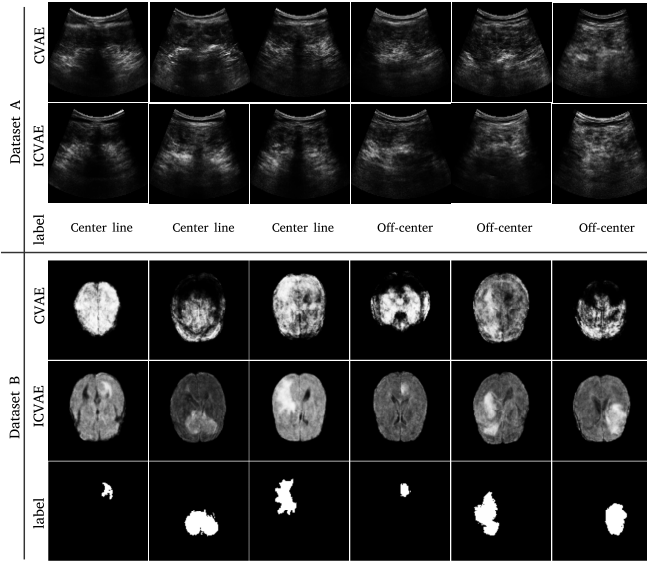


Fig. 8. Synthetic center line/off-center transverse US images of the spine and MR FLAIR planes with deformed tumor labels generated by CVAE and ICVAE after training on datasets A and B. Because of low dimensionality of the classification labels, both models are able to synthesize realistic transverse US images. However, since the latent space of ICVAE is encoding appearance features independent of the structures, it is capable of generating realistic brain MR planes.

was 3886.89, which indicates that the latent space has mainly encoded the input. However when trained on segmentation data (dataset B), the respective ratio was 5.70 that shows the latent variable is approximately equally dependent on the input and the label. Hence, this property of CVAE affects the performance of the model for image synthesis with modified labels for data augmentation. Figure 8 shows synthetic data generated by CVAE and ICVAE using deformed labels with the same samples from the prior $p(z)$ after training. Note that both models can generate realistic images given classification labels, but CVAE cannot generate realistic samples with modified segmentation labels.

The AdaTrain algorithm contributes to training of the discriminative model through 1) balancing the categories in dataset by incorporating prior knowledge and 2) adding more variance in the image space. The latter is achieved by adaptively adjusting the latent variable based on the performance of the discriminative model. In addition, more variance in the training data leads to more generalized models. This is apparent in Figure 6, which shows that the discriminators trained via AdaTrain (S3) have higher training loss value compared to the models of S1 and S2. However despite having higher training loss, the models indicated higher classification and segmentation performances on the unseen data (see Tables II and III).

The augmented data in each round of AdaTrain have regularized the models and therefore both classification and segmentation networks of S3 indicate higher accuracies on the test data, when compared against S1 and S2.

Although AUC is used as a measure of classification performance, it is a discrimination index that indicates the likelihood that one class has higher probability than the other, regardless

of the goodness of fit [80]. Hence, it is possible that a poorly fitted model can have a high AUC. According to table II, the baseline model indicated slightly higher AUCs, when trained on the original dataset versus statically augmented data (0.90 and 0.88 vs. 0.88 and 0.88). However since the training set is imbalanced, the model is biased *i.e.*, low sensitivity and high specificity, despite having a high AUC. This implies that although a cut-off threshold exists that yields good discrimination for the classification, the model is not properly fit and high AUC does not correspond to a more accurate model.

Although AdaTrain improves the accuracy of a discriminative model by leveraging the statistics of the data in conjunction with the prior knowledge about the dataset, model architecture is another key factor that has a significant impact on the accuracy. Therefore, an efficient network should be designed properly with enough learning capacity in order to be trained with the proposed adaptive data augmentation algorithm. This has been investigated via evaluating the performance of classification and segmentation models with state-of-the-art architectures using AdaTrain.

Lastly, the discriminative models, which are trained via the proposed augmentation method, provide “additional information” for clinical procedures to increase the confidence level of the operator. As operator experience with the system and interpretation of the anatomy grows, the operator is expected to be able to identify the outliers from the algorithm and simply recognize the target of interest.

V. CONCLUSION

In this study, we proposed an adaptive augmentation method for training discriminative models. In particular, we proposed a generative network based on variational auto-encoders that can learn the distribution of the input with respect to the label and the latent variable. The key difference of our model and the conventional conditional VAE is that we enforce independence between labels and the latent space. Therefore, we can adjust each of them independently for image synthesis. This is beneficial because by tuning the latent variable, we can control the characteristics of the image, while modifying labels enables one to incorporate prior knowledge about the dataset into the synthetic images. We showed that the performance of baseline discriminative models for both image classification and segmentation tasks can be improved by training with adaptively generating images based on the feedback from the discriminative model. In addition, the proposed method improved performance of discriminative models with more complicated architecture, namely, residual network and U-net. We validated this improvement with the use of ultrasound and MR data from test subjects for classification and segmentation.

APPENDIX A

The goal of the ICVAE is to learn a distribution $q(z|x)$, which approximates the true posterior $p(z|x)$. This is achieved by minimizing the KL divergence:

$$KL(q(z|x)||p(z|x)) = - \sum_z q(z|x) \log \frac{p(z|x)}{q(z|x)}$$

By conditioning $p(x, y, z)$ over x and substituting $p(z|x)$:

$$\begin{aligned}
 KL(q(z|x)||p(z|x)) &= - \sum_z q(z|x) \log \frac{p(x, y, z)}{p(y|z, x)p(x)q(z|x)} \\
 &= - \sum_z q(z|x) \log \frac{p(x, y, z)}{p(y|z, x)q(z|x)} \\
 &\quad + \sum_z q(z|x) \log p(x) \\
 &= - \sum_z q(z|x) \log \frac{p(x, y, z)}{p(y|z, x)q(z|x)} \\
 &\quad + \log p(x) \\
 \implies \log p(x) &= \sum_z q(z|x) \log \frac{p(x, y, z)}{p(y|z, x)q(z|x)} \\
 &\quad + KL(q(z|x)||p(z|x))
 \end{aligned}$$

Minimizing the KL divergence directly is not straightforward. On the other hand, $\log p(x)$ is independent of z , i.e., a constant. Therefore, we can maximize the first term of the right hand side of the above equation, which is referred to as the variational lower bound.

APPENDIX B

By conditioning $p(x, y, z)$ over z , the variational lower bound can be rewritten as:

$$\begin{aligned}
 \mathcal{L} &= \sum_z q(z|x) \log \frac{p(y|z, x)p(x|z)p(z)}{p(y|z, x)q(z|x)} \\
 &= \sum_z q(z|x) \log p(x|z) + \sum_z q(z|x) \log \frac{p(z)}{q(z|x)}
 \end{aligned}$$

The second term corresponds to the negative KL divergence between $q(z|x)$ and $p(z)$. $p(x, y, z)$ can be rewritten as:

$$\begin{aligned}
 p(x, y, z) &= p(x|y, z)p(y|z)p(z) \\
 &= p(y|x, z)p(x|z)p(z)
 \end{aligned}$$

and since $p(z) \sim \mathcal{N}(\mu, \sigma) \neq 0$ and y and z are independent, using the above equation, $p(x|z)$ can be written as:

$$p(x|z) = \frac{p(x|y, z)p(y|z)}{p(y|x, z)} = \frac{p(x|y, z)p(y)}{p(y|x)}$$

Therefore:

$$\begin{aligned}
 \mathcal{L} &= \sum_z q(z|x) \log p(x|z) - KL(q(z|x)||p(z)) \\
 &= \sum_z q(z|x) \log \frac{p(x|y, z)p(y)}{p(y|x)} - KL(q(z|x)||p(z)) \\
 &= \sum_z q(z|x) \log p(x|y, z) + \sum_z q(z|x) \log p(y) \\
 &\quad - \sum_z q(z|x) \log p(y|x) - KL(q(z|x)||p(z))
 \end{aligned}$$

Finally, since $\sum_z q(z|x) = 1$ and $\log p(y)$ and $\log p(y|x)$ are independent of z , the variational lower bound can be rewritten as:

$$\begin{aligned}
 \mathcal{L} &= \mathbb{E}_{q(z|x)}[\log p(x|y, z)] - KL(q(z|x)||p(z)) + \log p(y) \\
 &\quad - \log p(y|x).
 \end{aligned}$$

APPENDIX C

$\log p(Y) - \log p(Y|X)$ can be viewed as a Monte Carlo sampling error, because $p(Y|X)$ is given over a mini-batch, which is psuedo-randomly selected from $\mathbf{D}$ and $p(Y)$ is the true prior of the labels. Let μ_y and $\bar{y}$ be the mean of the true prior and the mean of a batch with size N , respectively. Therefore, $\bar{y} = 1/N \sum_{i=1}^N y_i$. Since each batch is selected randomly from the dataset, the expected value of y_i is μ_y . Therefore the expectation of the error ($\mu_y - \bar{y}$) is:

$$\begin{aligned}
 \mathbb{E}[\mu_y - \bar{y}] &= \mu_y - \mathbb{E}\left[1/N \sum_i y_i\right] = \mu_y - 1/N \sum_i \mathbb{E}[y_i] \\
 &= \mu_y - N \times 1/N \mu_y = 0.
 \end{aligned}$$

Since there is no bias in selecting the batches during training, the expected value of the error is zero. However, the error has a non-zero variance, which depends on the size of the batch. This can be shown by taking the variance of $\bar{y} - \mu_y$, which is:

$$V(\bar{y} - \mu_y) = V\left(1/N \sum_i y_i\right) - 0 = 1/N^2 V\left(\sum_i y_i\right).$$

Since samples in a batch are independent, the variance of their sum equals the sum of their variance. Therefore, $V(\bar{y} - \mu_y) = 1/N^2 \sum_i V(y_i)$, which implies that $V(\bar{y} - \mu_y) = \frac{1}{N^2} NV(y)$. $V(y)$ is the variance of all labels. Since $\mathbb{E}[\bar{y} - \mu_y] = 0$ and $V(\bar{y} - \mu_y) = V(y)/N$, the central limit theorem suggests that the error would have a normal distribution with mean 0 and standard deviation $\sqrt{V(y)/N}$, if N is large enough ($\mathcal{N}(0, \sqrt{V(y)/N})$). Notice that a large N would reduce the standard deviation of the Monte Carlo sampling error, and accordingly, the difference between the distributions of the randomly selected sample and the true prior would be negligible.

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